Noise-space RL for arm–hand coordination in a frozen SignSparK

Final results: ACMS multi-seed replication and the FG-1 feasibility gate

How2Sign (ASL) · 7 November 2026

Released-checkpoint result, not a paper reproduction · motion statistics only

Bottom line

+0.0022
pooled Δ right arm–hand coordination, 4 new seeds

seed-sensitive
registered replication category

≈1/23
of the registered 0.05 relevance threshold

all pass
pre-registered preservation checks, 5 runs

- Pooled 95% CI across runs +0.0022 [-0.0013, +0.0056] includes zero; $I^2 = 0.97$.
- In the FG-1 fresh-noise gate: learned M-C negative (held-out Δ +0.0034); per-clip ACMS inconclusive (held-out Δ +0.1016); bilateral SC-MC positive (held-out Δ +0.0893).
- Conclusion: the amortised RL policy gives no practically relevant gain; per-clip control is feasible.

The problem: generated hands ignore the arms

- SignSparK samples body and each hand from independent noise in gloss-free text-to-pose generation.
- Arm–hand coordination in generated How2Sign samples ≈ 0.02 vs ≈ 0.45 in references (chance level).
- Goal: reduce this deficit with RL on top of the frozen released checkpoints, without changing per-stream motion.
- Out of scope (user decisions): fine-tuning, test split, human evaluation, perceptual or semantic claims.

Formulation: a contextual bandit over the initial noise

- Action: initial noise of the right-hand stream; one decode per episode; terminal reward.
- Per-frame MDPs rejected: the 10-step rectified-flow generator is non-autoregressive.
- Reward: −|coord_ah^R − 0.4514| (frozen train-reference target, metric R v1.1).
- Constraints (articulation, jerk, extremes, activity) by dual ascent; λ stayed 0 in all runs.
- Primary statistic: paired clip-mean Δ trained − uncoupled on dev-256 × 4 shared noises.

ACMS: arm-conditioned temporal mean shift

- $\varepsilon_R = \xi + \mu_\phi(\hat{s})$, $\mu_\phi = \beta \cdot U \cdot \tanh(W * \hat{s} + c)$; $\hat{s}$ = generated arm speed profile.
- Rank 4, 17-tap kernel, 433 parameters; KL = $\frac{1}{2}\|\mu\|^2$ per frame < 1 nat/frame by construction.
- REINFORCE with leave-one-out baseline (G = 4), Adam 0.01, clip 1.0, 125 updates / 500 episodes.
- Stage 5 run 7000: Δ = +0.0022 [+0.0014, +0.0031], pre-registered verdict positive but negligible.

Protocol and safeguards

- Seeds 7001–7004 vary init, clip sampling (5000 + d) and training noise (50M + d·1M) independently; eval noise shared.
- Runs = unit of inference: per-run CIs, t-interval across runs, clip and hierarchical bootstrap, Q / I^2 / τ^2.
- Category rule fixed before running; update 125 primary; no best-run or best-checkpoint selection.
- Checkpoint SHA-256 before/after; Stage 4/5 artifacts hash-snapshotted; TRAIN/DEV only, test split never opened.
- Metrics, validity rules, targets, rewards and thresholds unchanged.

1 · ACMS multi-seed replication

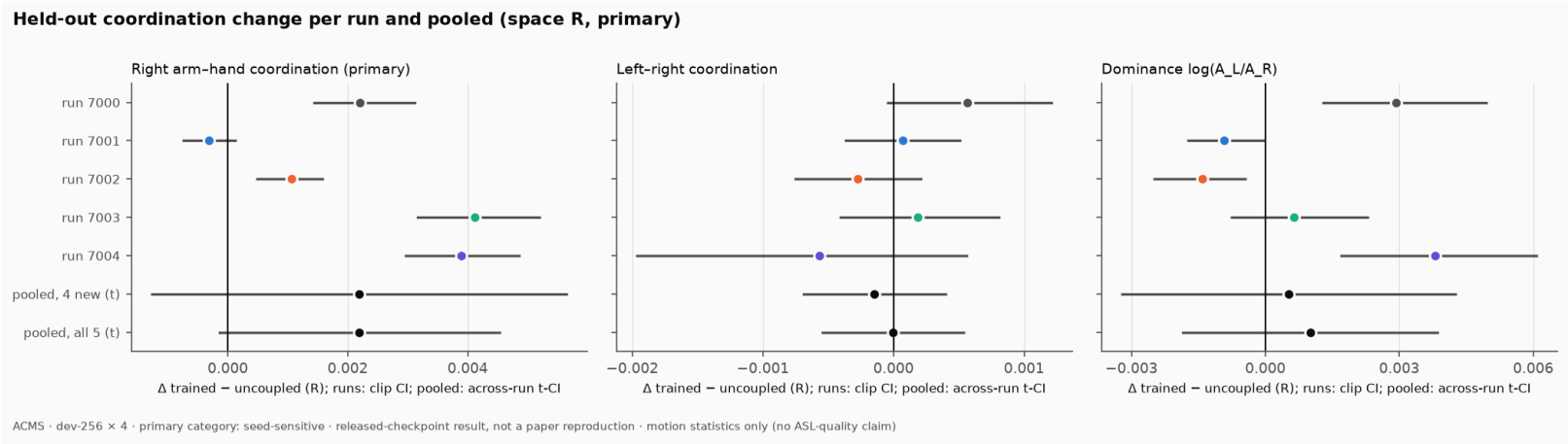
Four new seeds, original configuration, frozen SignSparK

Per-run results (dev-256 × 4, space R)

Run	Δ coord_ah^R [95% clip CI]	Verdict	KL mean	Train s
7000	+0.0022 [+0.0014, +0.0031]	positive	0.0037	284
7001	-0.0003 [-0.0007, +0.0001]	negative	0.0026	283
7002	+0.0011 [+0.0005, +0.0016]	positive	0.0029	282
7003	+0.0041 [+0.0032, +0.0052]	positive	0.0035	273
7004	+0.0039 [+0.0030, +0.0049]	positive	0.0088	273
Pooled 4 new (t)	+0.0022 [-0.0013, +0.0056]	seed-sensitive		
Pooled all 5 (t)	+0.0022 [-0.0001, +0.0045]	seed-sensitive		

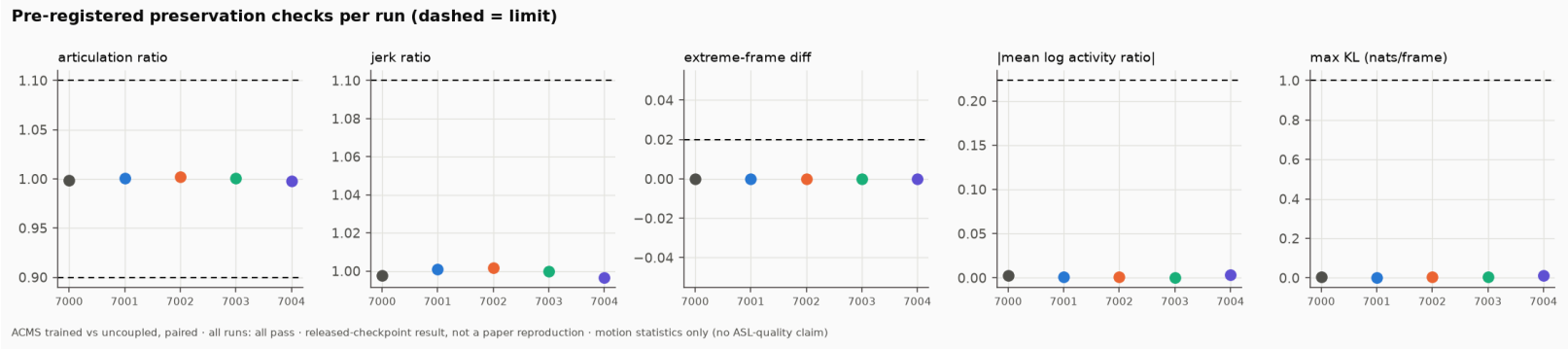
- All runs completed; weights unchanged; 0 undefined statistics; uncoupled samples bitwise identical across runs.

Effect per run and pooled



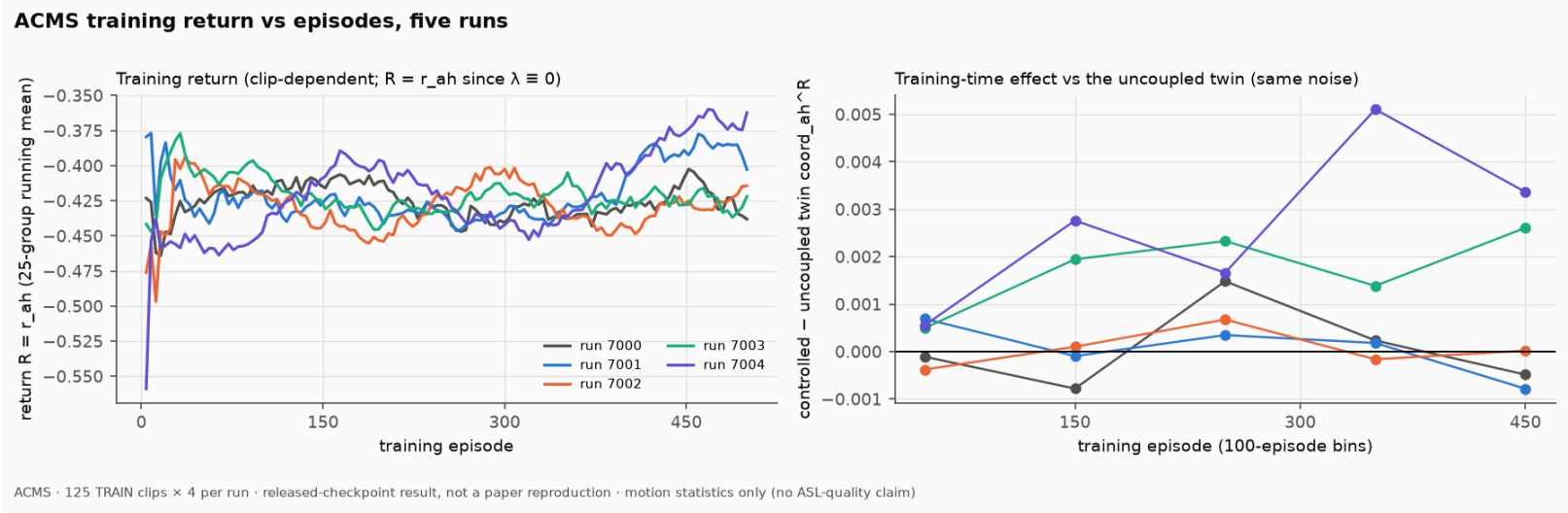
- Pooled 4 new +0.0022 [-0.0013, +0.0056]
- Clip bootstrap +0.0022 [+0.0018, +0.0026] (eval sampling only)
- $I^2 = 0.97$; Q = 108 (df 3)
- Dominance and L–R: no consistent direction

Preservation: nothing else changed



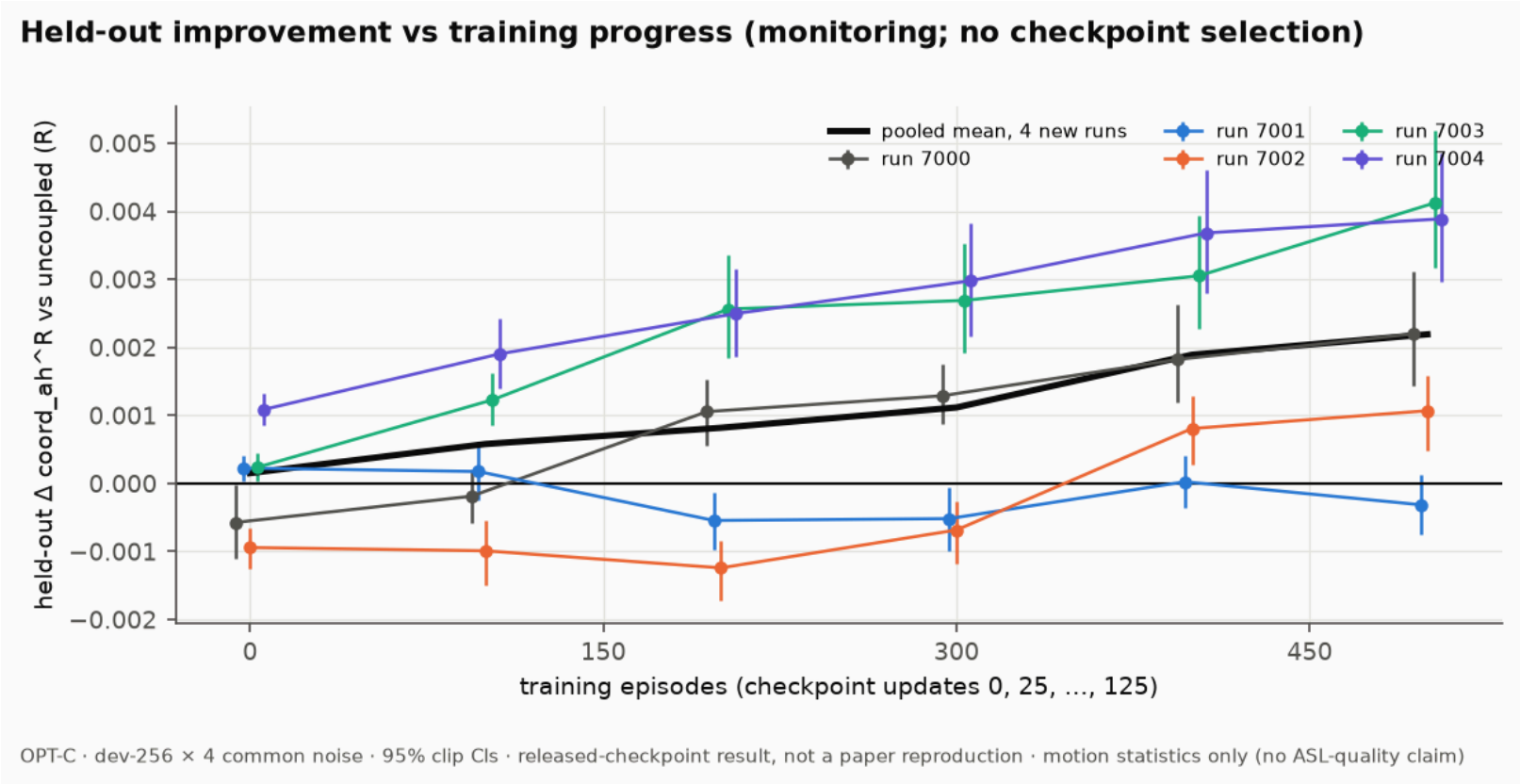
- Articulation, jerk, extremes, activity, KL: all pass in all runs
- Ratios within 0.5% of 1
- Max KL 0.0114 nats/frame

Training dynamics



- Return is clip-dependent and noisy
- Training-time controlled – twin effect small in all runs
- No constraint ever active

OPT-C: held-out effect across checkpoints (monitoring)



- Updates 0, 25, 50, 75, 100, 125 evaluated (R only)
- Update 125 stays primary; nothing selected

Stability across seeds



- Effect, dominance, L–R, activity, KL and β per run
- Spread between seeds dominates evaluation noise

2 · FG-1 feasibility gate

Per-clip, like-for-like, fresh noise

FG-1 design

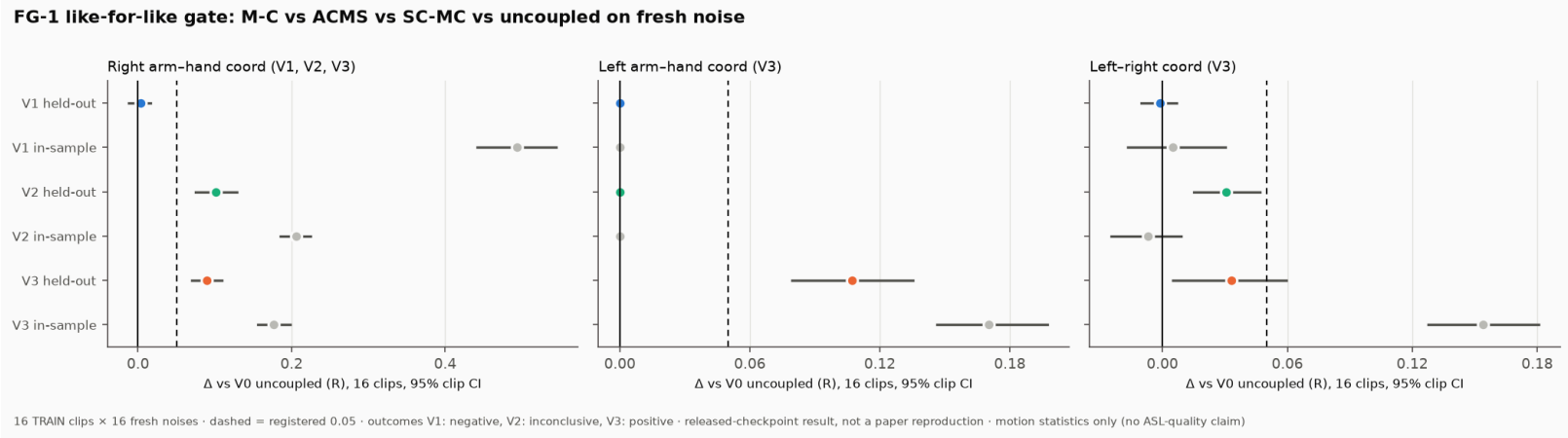
- 16 TRAIN clips (8 short, 8 long), disjoint from ACMS training clips; 16 fit + 16 held-out noises per clip.
- V0 uncoupled · V1 learned M-C coupling · V2 per-clip ACMS · V3 bilateral SC-MC (new direction).
- Pathwise gradients through the frozen unrolled sampler; Adam 0.05, 30 iterations.
- Positive: CI lower > 0, point ≥ 0.05, preservation passes · Negative: CI upper < 0.05 · else inconclusive.
- Operational decisions D1–D10 documented before running (e.g. V3 cap 120 min); protocol unchanged.

FG-1 results (right arm–hand, space R)

Variant	Params/clip	Held-out Δ vs V0 (R, right)	In-sample Δ (last iter)	Gap	KL max	Preserv.	Outcome
V1	9060	+0.0034 [-0.0121, +0.0166]	+0.4938 [+0.4418, +0.5447]	+0.4904	2.980	pass	negative
V2	433	+0.1016 [+0.0753, +0.1288]	+0.2058 [+0.1849, +0.2245]	+0.1042	0.119	FAIL	inconclusive
V3	866	+0.0893 [+0.0703, +0.1091]	+0.1765 [+0.1559, +0.1983]	+0.0872	0.112	pass	positive

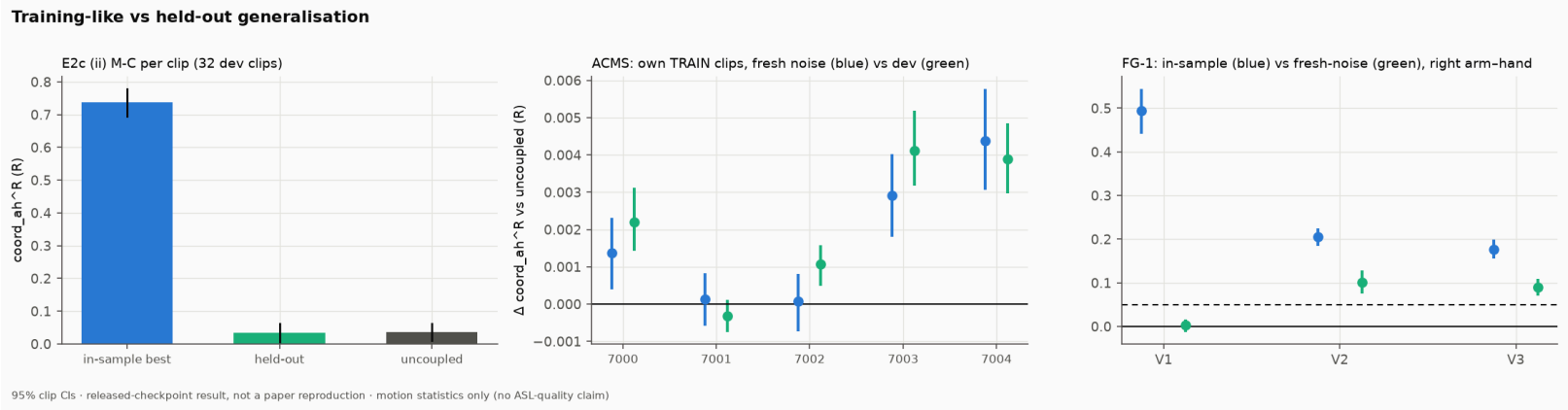
- In the FG-1 fresh-noise gate: learned M-C negative (held-out Δ +0.0034); per-clip ACMS inconclusive (held-out Δ +0.1016); bilateral SC-MC positive (held-out Δ +0.0893).

FG-1: in-sample vs held-out



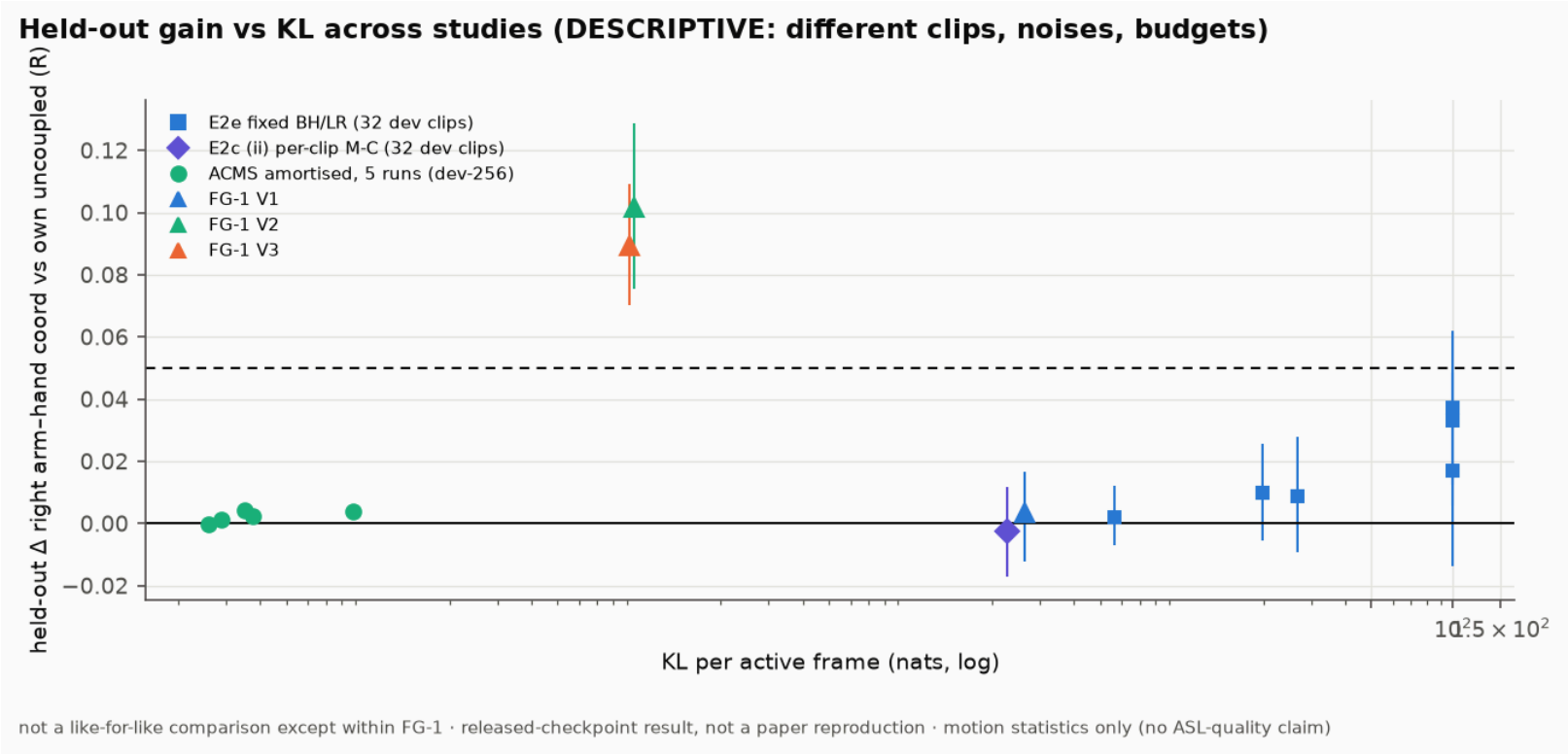
- Grey: in-sample (fit noises)
- Coloured: held-out fresh noise
- Dashed: registered 0.05

Generalisation across studies



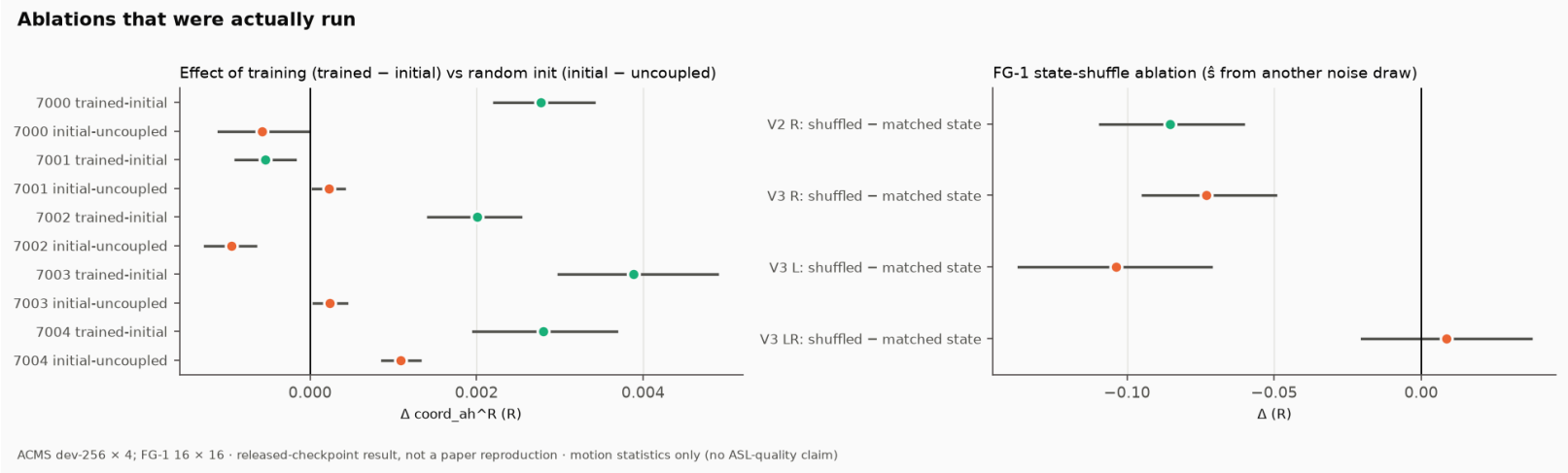
- E2c (ii): in-sample 0.74 vs held-out ≈ uncoupled
- ACMS: own TRAIN clips vs dev
- FG-1: in-sample vs fresh noise

Gain vs KL across studies (descriptive)



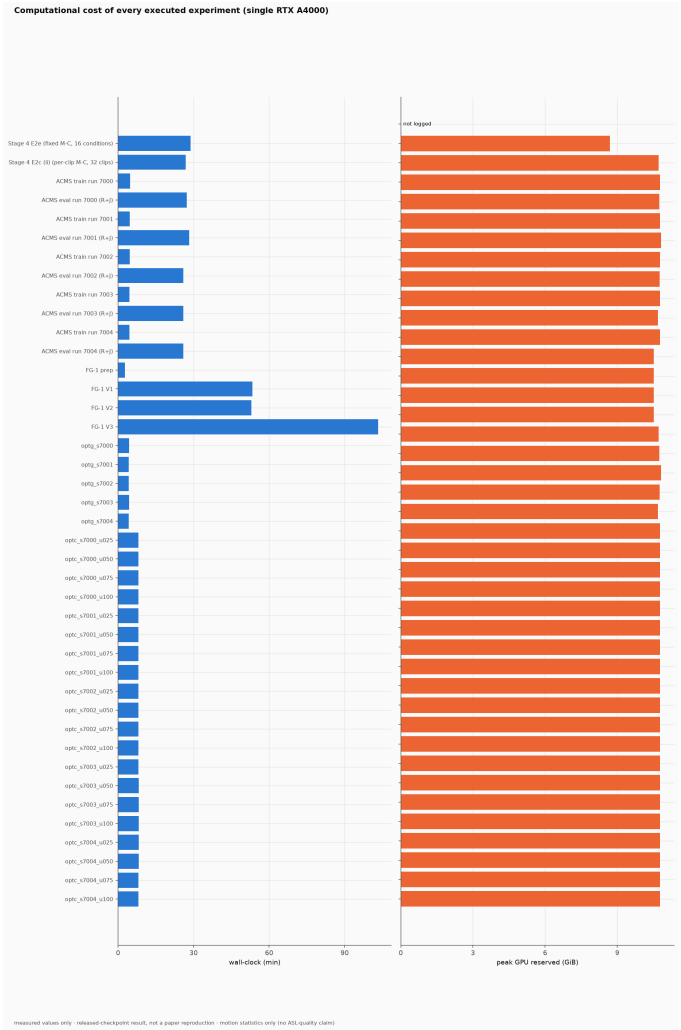
- Different clips, noises, budgets
- Only FG-1 is like-for-like

Ablations actually run



- Trained − initial vs initial − uncoupled (ACMS)
- State shuffle: ŝ from another noise draw (FG-1 V2/V3)

Compute (single RTX A4000 16 GB)



- ACMS training ≈ 5 min, evaluation ≈ 26–28 min per run
- No GPU rental; jobs run sequentially

Consolidated results

Experiment	Unit / n	Primary statistic (R)	Estimate [95% CI]	Outcome
Stage 4 E2e fixed M-C coupling	16 conditions, 32 dev clips	Δ arm–hand (T3)	R 0.017, L 0.022 (CIs incl. 0)	FM13 stop (negative)
Stage 4 E2c (ii) per-clip M-C	32 dev clips	held-out gain	-0.003 [-0.017, +0.012]	negative (FM4)
Stage 5 ACMS, run 7000	dev-256 × 4	Δ trained − uncoupled	+0.0022 [+0.0014, +0.0031]	positive, negligible
ACMS multi-seed (7001–7004)	4 runs	pooled Δ (t)	+0.0022 [-0.0013, +0.0056]	seed-sensitive
ACMS all five runs (secondary)	5 runs	pooled Δ (t)	+0.0022 [-0.0001, +0.0045]	seed-sensitive
OPT-G train-like generalisation	4 new runs	pooled Δ on own TRAIN clips	+0.0019 [-0.0015, +0.0053]	monitoring
FG-1 V1	16 TRAIN clips × 16	held-out Δ vs V0	+0.0034 [-0.0121, +0.0166]	negative
FG-1 V2	16 TRAIN clips × 16	held-out Δ vs V0	+0.1016 [+0.0753, +0.1288]	inconclusive
FG-1 V3	16 TRAIN clips × 16	held-out Δ vs V0	+0.0893 [+0.0703, +0.1091]	positive

Metrics not reported, and why

Metric	Status	Reason
DTW-JPE / MPJPE vs. reference as success measure	not used	favours collapse in GF (static mean beats samples; M5–M6)
BLEU / back-translation	not used	within-text uncorrelated with pose error; hackable by selection (R6)
FID / diversity / multimodality	unavailable	no validated ASL feature extractor
Task success rate	unavailable	no episodic task or success criterion exists
Regret	unavailable	no oracle policy value is defined
PPO / GRPO / SAC comparison	unavailable	never run; no comparative claim made
Perceptual naturalness, intelligibility, ASL quality	out of scope	human evaluation removed (rev. 4.0)
How2Sign test split	never opened	only registered TRAIN and DEV used

- Earlier slide deck not in workspace: deck-to-deck coverage verification unavailable.

Limitations

- Four new runs: wide across-run interval; seed-sensitive reflects heterogeneity and low power.
- R coordination is validated automatically only; no perceptual validity.
- J includes SMPL-X skinning leakage (~1–2%); descriptive only.
- FG-1: 16 TRAIN clips, one fit each, pathwise gradients; controllability, not an RL policy.
- V2 misses the articulation bound narrowly (1.102 vs 1.10); applied as registered.
- Checkpoint pretraining split usage undocumented (absolute levels only).

Conclusion

ACMS gain +0.0022 pooled, seed-sensitive, ≈1/23 of the threshold

In the FG-1 fresh-noise gate: learned M-C negative (held-out Δ +0.0034); per-clip ACMS inconclusive (held-out Δ +0.1016); bilateral SC-MC positive (held-out Δ +0.0893).

Per-clip control works on fresh noise; one shared RL policy does not. Project closed on this evidence.